

Impact of Collaborative Virtual Environments on Design Process

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Abstract. Collaborative Virtual Environment (CVE) is a form of environments where multiple users, whether remote or not, can collaboratively develop and control virtual 3D models, using advanced information and communication technology (ICT). CVE is expected to have significant impacts on design and construction of civil and built environments, as the following: CVE will contribute to improve the quality of civil and built environments, and decrease development period and project cost. CVE is expected to design more environment-friendly, green buildings and civil infrastructure by analyzing environmental aspects in the early stage of design. More creative design is also expected in CVE by the collaborative atmosphere among heterogeneous experts. CVE will contribute to obtaining social acceptance of design from citizens and stakeholders through better presentation and communication. CVE is expected to give significant impact on the reform of business processes of design and construction from design-bid-build to design-build.

Keywords: concurrent engineering, product model, project process, design-build, integration.

1 Introduction

To design civil and building environments is a creative, complicated and advanced activity. One expert usually cannot design the entire thing and many experts in various disciplines participate in the design process (Luo and Dias, 2004). The traditional design adopts a methodology that divides the design object into foundations, structures, facilities, aesthetics, etc. and that classifies design as preliminary, detail, construction, etc. Participants in designing civil and built environments include owners, architects, structural engineers, foundation engineers, facility engineers, construction engineers, governmental officers, other stakeholders, etc. Each expert executes his or her own task of the discipline and sends the results to the owner or other experts on the basis of contracts. This procedure is error-prone and tends to take much time because other experts have to re-enter the data generated by previous experts into their computers. If a problem occurs in the later phase, designer may have to return to former phase for modification, ending up

with significant cost increase. Moreover, knowledge and experience in the downstream phase may not be utilized in the upstream phase, which could hinder the opportunity of modifying and enhancing previous design for the future.

The environment and energy issue is one of the most crucial items to tackle in the 21st century. Civil and built environments spend a large amount of fossil fuels and emit much carbon dioxide. In the design of buildings and infrastructure, environmental aspects should be investigated and incorporated in the early stage of design, especially for designing energy-efficient environments. However, environmental aspects are usually analyzed in the later stage and treated like adjuncts rather than key roles.

Design and construction of civil and built environments often confront claims by citizens and other stakeholders, mainly due to misunderstanding during the design phase. Such troubles may stop the project and could lead to a severe damage to the project developers.

2 Concurrent Engineering

In mechanical design field, 3D CAD systems have begun to be used since 1970s and CAD was merged with Computer-Aided Manufacturing (CAM) and Computer-Aided Engineering (CAE), which lead to Computer-Integrated Manufacturing (CIM). In CIM of automobiles, bodies and main parts are designed using CAD, then the geometric and property data is transferred to structural analysis software packages based on the finite element method (FEM), and various simulations are performed. The design is then modified based on the result of simulations. After the design is complete, steel plates are cut and drilled using numerical control (NC). Since software packages such as CAD, FEM, simulations, NC, etc. were developed by different companies and researchers, the interoperability of data had become an important issue. In early 1980s, the object-oriented paradigm was born, and university researchers and industry engineers began to develop common data models for representing products, i.e., product models. Some of such efforts have become parts of Standard of the Exchange of Product model data (STEP) of International Organization for Standardization (ISO) (Ota and Jelinek, 2004).

As the model-based method of design and manufacturing spread in industry, the concept of concurrent engineering emerged. In concurrent engineering, an object to-be-designed is divided into several parts, or is divided into several design tasks, which are executed by each expert or group of experts separately and concurrently. Then, the experts meet and merge the results of the design. Based on the review of the design, they execute the next tasks separately, and this process is repeated until the design is done. Concurrent engineering has been expected to get the design done faster and in a shorter time than the traditional procedural design. Further, since multiple experts in heterogeneous disciplines gather, discuss freely and generate new ideas, concurrent engineering has been expected to generate more creative design (Vivasqua and de Souza, 2004).

However, while experts are doing their own tasks, they do not know what others are doing. Thus, when they show their results each other at the meeting, problems such as conflicts of design objects usually occur and they have to solve the problems by discussion. Sometimes some of the design results may be wasted. As values, knowledge, and terms of experts in various disciplines are often different, they tend to argue without full understanding, have difficulty in communication, and cumulate frustration. Although they should discuss with other experts during the parallel work period if they encounter a trouble or waver in decision-making, they tend not to. Actually, although concurrent engineering looked promising then, it is not so easy in practice.

3 Integration

Design of civil and built environments varies according to the phase from preliminary design, detailed design, to construction design. Since civil and architectural structures are built in natural or urban environment, all the conditions or specifications cannot be given a priori at the beginning of design. Much information is collected by surveying, site investigation, etc. as the design proceeds. Thus, design is often modified or sometimes forced to change itself and the amount of information of the design becomes larger and larger in the course of development.

In the design of civil and built environments, requirements such as safety, serviceability for different types of users, aesthetics, impact for nature and environment, cost, schedule, etc., must be investigated and satisfied. In addition, adjunct facilities and machines have to be designed. Therefore, many experts in heterogeneous disciplines execute investigation in each area. The amount of work and the level of accuracy are different according to the design stage. Most of the design tasks are done using computer software packages. As in the mechanical design area, the interoperability of data among software packages was low, development of product models for buildings started in 1990s. Now the de facto standard of building product models is Industry Foundation Classes (IFC).

Since around 2005, the term Building Information Modeling (BIM) has been spread in building industry. BIM has not been clearly defined yet, but it means a technology for executing design of buildings and civil infrastructure quickly and efficiently by exchanging and sharing design information based on the product models such as IFC that integrates heterogeneous application software systems. The International Alliance for Interoperability, which is in the process of changing its own name to buildingSMART, has organized events called BIM-Storm recently. In BIM-Storm, each participating group consists of several design experts in heterogeneous disciplines such as architecture, structural design, facility management, construction management, etc. The groups compete to design an assigned building by exchanging and sharing design information based on IFC via the Internet in only a few days. In Japan, a similar event called Build Live Tokyo 2009 was organized and six groups participated in it. They designed a virtual building in a certain place in Tokyo within 48 hours. Many experts in building industry are trying to change the design process significantly by the concept of BIM.

4 Project Processes

The process of design and construction of civil and built environments, usually of single part production, is considerably different from that of products manufactured and mass-produced at factories. Especially, as for public civil infrastructure, design and construction are separately ordered by governmental, municipal and other public agencies, and the same private company is not able to receive both design and construction orders in principle. Because if one company performs both design and construction, the structure may be designed unnecessarily larger or more luxurious so that the construction cost becomes higher than normal, which is a serious concern for clients or owners. Although the design and construction separate ordering method, i.e., design-bid-build is effective for preventing corruption, it is very difficult to adopt the concurrent engineering method among design consultants and general contractors unlike manufacturing industry. Thus, consultants' design is not usually reflected by knowledge and experience of contractors, which may lead to various problems during construction. Further, if consultants know more new construction methods created by contractors, better and cost reduced design may be possible. Sharing and using knowledge and experience of construction by contractors is increasingly important and effective in the design phase.

In the United States and some other countries, Construction Manager (CM) with a wealth of knowledge and experience in design, construction, and management works among owners, design consultants, and contractors for managing a construction project smoothly from planning to complete of construction. In Japan, although CM has not been widely adopted yet, design-build method, which is commonly used in private building projects, is now under consideration for applying to public civil engineering works. Furthermore, new project methods such as Build-Operate-Transfer (BOT), Private Finance Initiative (PFI), Public Private Partnership (PPP) are investigated for public works. These new project methods are expected to break the limitations of the traditional design and construction separate ordering method.

5 Social Acceptance of Design

Design of civil and built environments is not satisfactory if it is not accepted by the society yet. The design becomes realistic only if it is understood and accepted by citizens and other stakeholders, i.e., society. However, even if the designed project is thought to be understood by the citizens and stakeholders, once the construction is started, they may think that the situation is different from what they thought. Because professional documents such as 2D drawings and reports describing the project tend to be hard to fully understand for novice people. The gap between their understanding and the reality could lead to a serious social trouble or problem that may end up re-design, postponement or cancellation of works.

Thus, when government, public agencies, or private developers perform urban planning, civil and building engineering projects, they often develop 3D computer

graphics (CG) models representing the urban areas, natural environments, etc. and present them to the citizens and other stakeholders at public hearings, using virtual reality (VR) software. Figure 1 shows snapshots of VR presentation of light rail transit (LRT) project planned in Sakai, Osaka, Japan (Yabuki et al., 2009).

At public hearings or explanation meetings, VR tends to be used just for presenting planned or designed projects in an easy-to-understand manner. However, Collaborative Virtual Environments (CVE) could change presentation and discussions at public hearings so that developers can modify the 3D VR model according to the request of stakeholders and present what-if scenarios interactively



Fig. 1. Snapshots from presentation of the LRT project in Sakai, Osaka, Japan.

6 Collaborative Virtual Environments

Computer technology is fast evolving. Concurrent engineering, CIM, CSCW, etc. develop into CVE by the evolution of 3D/4D CAD and VR technologies. CVE is a form of environments where multiple users, whether remote or not, can develop and control virtual 3D models with immersive feeling, communicating and collaborating each other. For example, an immersive VR dome facility named CyberDome of Panasonic Electric Works has a dome of 8.5m in diameter, and 18 projectors are used to display 3D stereoscopic VR images and videos. Thirty people can participate in the dome simultaneously.

In the research field, not only visualization but also multi-modal CVE systems are being developed. Multi modal includes not only vision, smell, taste, listening,

and touch but also input device such as 3D mouse, voice recognition, haptic feedback (Di Fiore et al., 2004).

The central part of CVE is a product model, which enables users to share and exchange data of products, processes, organizations, etc. Utilizing CVE, owners, designers, contractors, and other experts will collaboratively share and exchange design data. They will concurrently execute and compare a number of design alternatives, and will explain the plan and design to citizens and other stakeholders and negotiate immediately with them, considering the impact to cost and other conditions.

7 Impacts of CVE

CVE is expected to give significant impacts on design and construction of civil and built environments. The following sub sections describe impacts of CVE on design and construction.

7.1 Quality, Time and Cost

CVE will contribute to decrease of mistakes in design and construction, which will be effective for improvement of quality and decrease of accidents. Many mistakes in design and construction tend to occur due to misunderstanding in generating and reading 2D drawings and in communication among participants in design. CVE gives 3D VR world that designers and engineers can see and feel immersed in it, so that they can detect mistakes easily.

As participants for design, even if they are remote, can communicate with ease in CVE by video conferencing, decisions can be made more quickly and thus, wasted design and investigation will be decreased, which will reduce the period of design and construction and project cost.

7.2 Environmental Aspects

CVE is expected to contribute to realize the design which environmental aspects are more emphasized. As mentioned before, in current design of buildings and structures, most environmental analyses are done after basic design decisions are already made. It has been quite difficult to change the design significantly for improving energy saving, for decreasing the emission of carbon dioxide, and for giving less impact to environment. In CVE, environmental engineering experts will be able to participate in design from the early stage and suggest better design.

7.3 Creative Design

CVE is also expected to contribute to generate more creative design. Basic design done by an architect alone usually has limitations, while collaboration among several heterogeneous experts tends to generate more original ideas. Interesting and creative ideas are likely to be blocked in mind of designers because such ideas

look unrealistic or too expensive or fragile. However, if experts in other fields get to know the idea, they can investigate the feasibility immediately in CVE.

7.4 Social Acceptance

CVE will contribute to better presentation and communication at public hearings and mutual understanding by citizens, stakeholders, designers, engineers, and owners. As hardware and software of CVE evolve continuously, public agencies and designers will soon be able to perform what-if analysis corresponding to the request from participants at public hearings and compare multiple virtual worlds concurrently. Current public hearings generally take place at halls. Future public hearings will take place more and more virtually and remotely via the Internet in the CVE, where citizens will ask questions, give opinions, criticize some points, etc. and design experts respond to them.

It takes considerable time and cost to develop a 3D VR urban model covering wide area. CVE does not necessarily mean 3D VR model but it may be a mixture of real and virtual worlds such as Mixed Reality (MR) and Augmented Reality (AR). MR/AR will be an effective means to realize CVE.

7.5 Business Process Reform

CVE is expected to give significant impact on the reform of business processes of design and construction of civil infrastructure and buildings. Current design-bid-build method is not collaborative. Each project participant does his or her own task alone based on the contract and specification. This process is approaching the limits of its performance. On the other hand, CVE will encourage project participants to meet and communicate with each other more than now. They can share 3D models of civil and built environments and investigate them with their software without much frustration. Then, they will realize what hinders the design process is not the technology any more but it is the business process, which is based on many, rigid, separate contracts.

New technology such as CVE and reform of business process are like two wheels of an automobile, which must work together inseparably. Old business process should be reformed so that new information and communication technology (ICT) will use its abilities to the full.

The author feels that design-build contract will be one of the most promising business processes for future public infrastructure design and construction, considering the affinity for CVE. Corruption in design-build must be banned. The third party's review and suggestions may be a solution to regulate and control the design utility.

We should identify problems and concerns in promoting CVE and business process reform and solve them for sustainable development and better quality of life.

8 Conclusion

In this chapter, current design and engineering processes were reviewed and issues and problems in design and construction of civil and built environments were identified. Current innovation in ICT and efforts in application of advanced VR and integration technologies evolve the traditional design environments into CVE. CVE is expected to give the following significant impacts on design and construction of civil and built environments.

- CVE will contribute to improve the quality of civil and built environments, decrease development period and project cost.
- CVE is expected to design more environment-friendly, green buildings and civil infrastructure by incorporating environmental analysis in the early stage of design.
- More creative design is also expected in CVE by the collaborative atmosphere among heterogeneous experts.
- CVE will contribute to obtaining social acceptance of design from citizens and stakeholders through better presentation.
- CVE is expected to give significant impact on the reform of business processes of design and construction from design-bid-build to design-build.

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